

GitHub Copilot CLI, VS Code & Power BI Skills

A beginner-friendly, step-by-step setup and validation guide •
Windows-first with macOS alternatives • Preview

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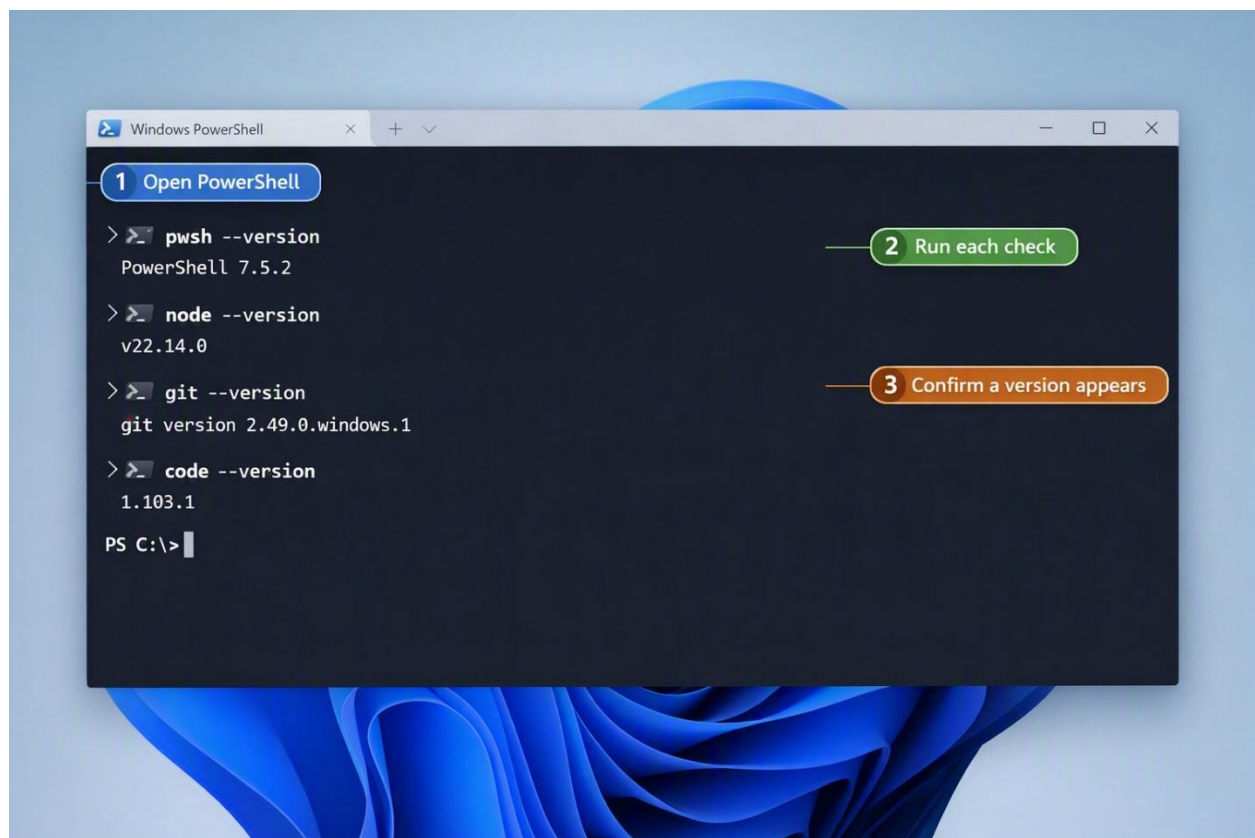
Outcome: In about 30–45 minutes, you will install the required tools, sign in to GitHub Copilot, add Microsoft’s Power BI authoring skills, connect the CLI to VS Code, and run a safe first test. The Power BI agentic authoring experience is currently in preview, so review generated PBIR and DAX before using them in production.



Figure 1. Illustrated setup flow. Use the copy/paste commands in this guide as the source of truth; screen labels and versions can vary by release.

1. Before you begin

Requirement	What you need	How to check
GitHub account	Copilot access (Free or paid); organizational policy must allow Copilot CLI	Sign in to GitHub and confirm Copilot access
PowerShell	Windows: PowerShell 7 or later	<code>pwsh --version</code>
Node.js	Node.js 22 or later for the current npm-based CLI path	<code>node --version</code>
Git	Current Git client	<code>git --version</code>
Power BI	Power BI Desktop recommended; permission to the workspace and content you will use	Open Desktop and confirm workspace access
VS Code	Current stable release	<code>code --version</code>



Screenshot 1. Check each prerequisite in PowerShell. Your version numbers may be newer.

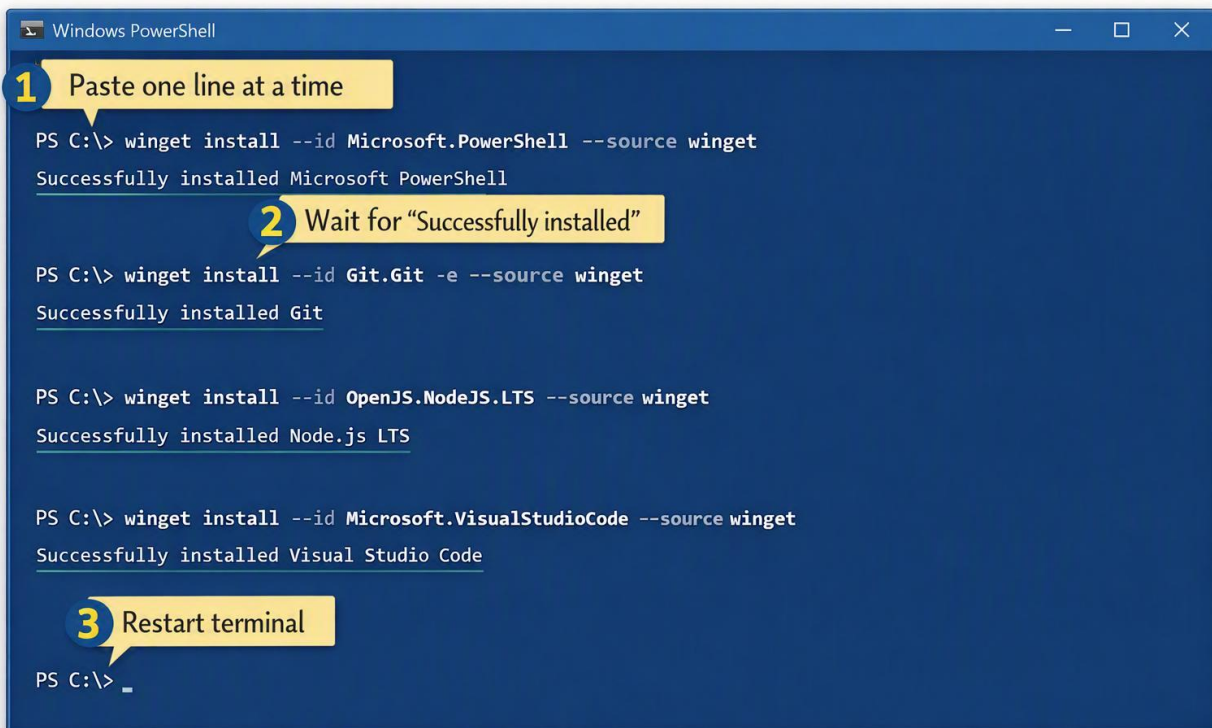
Important distinction: Power BI Agent is not a switch inside Power BI. It is a bundle of agent skills plus tools installed into an AI coding agent. The recommended Power BI setup is the separate powerbi-authoring plug-in; the broader fabric-skills bundle is optional when you also need Fabric workload skills.

2. Path A — Set up GitHub Copilot CLI

Step 1 — Install the base tools

Windows (PowerShell)

```
winget install --id Microsoft.PowerShell --source winget
winget install --id Git.Git -e --source winget
winget install --id OpenJS.NodeJS.LTS --source winget
winget install --id Microsoft.VisualStudioCode --source winget
```



Screenshot 2. Paste one WinGet command at a time, wait for success, and restart the terminal.

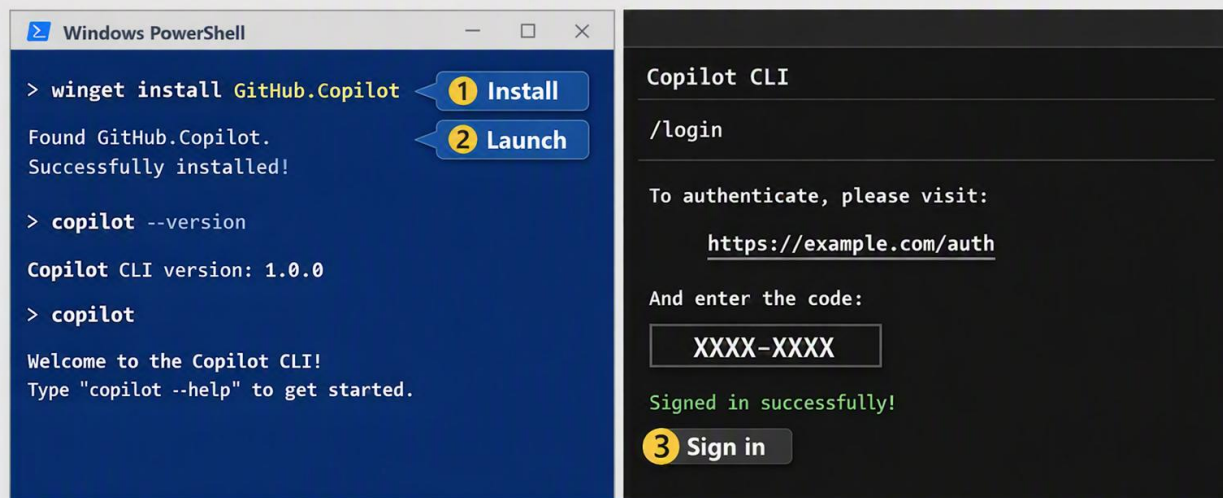
macOS: Install Node.js and VS Code with Homebrew: `brew install node` and `brew install --cask visual-studio-code`. Git is included with Apple developer command-line tools or can be installed through Homebrew.

Step 2 — Install and sign in to Copilot CLI

```
# Preferred Windows package
winget install GitHub.Copilot

# Cross-platform npm alternative
npm install -g @github/copilot

# Verify and launch
copilot --version
copilot
```



Screenshot 3. Install and launch Copilot CLI, then complete browser authentication. The pictured code is intentionally fictional.

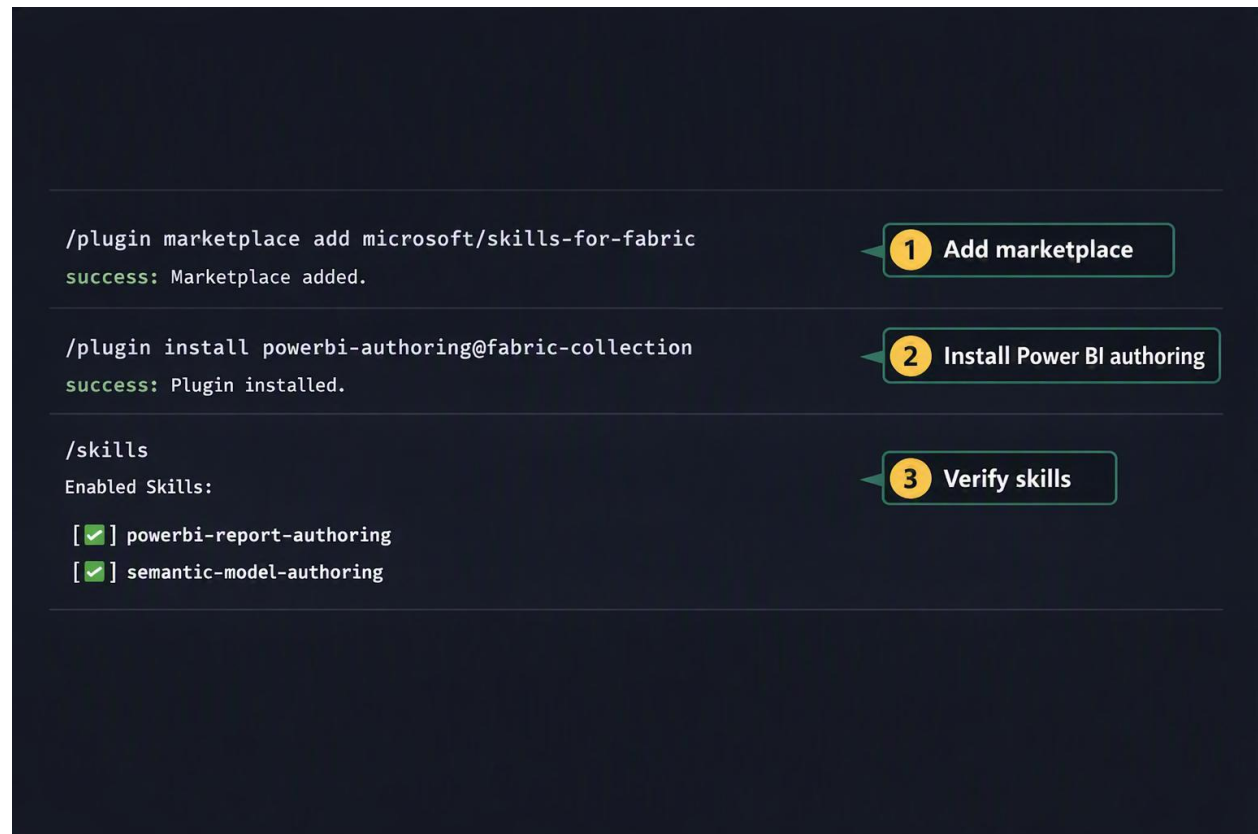
At first launch, enter `/login` if prompted, choose the correct GitHub account, complete browser authentication, and return to the terminal. If your organization provides Copilot, its administrator must allow Copilot CLI.

Step 3 — Install Power BI Skills

```
# Run these inside the interactive Copilot CLI session
/plugin marketplace add microsoft/skills-for-fabric
/plugin install powerbi-authoring@fabric-collection
```

```
# Optional: add the complete Fabric bundle too
/plugin install fabric-skills@fabric-collection

# Restart, then verify
/quit
copilot
/skills
```



Screenshot 4. Add the Microsoft marketplace, install Power BI authoring, and confirm both authoring skills are enabled.

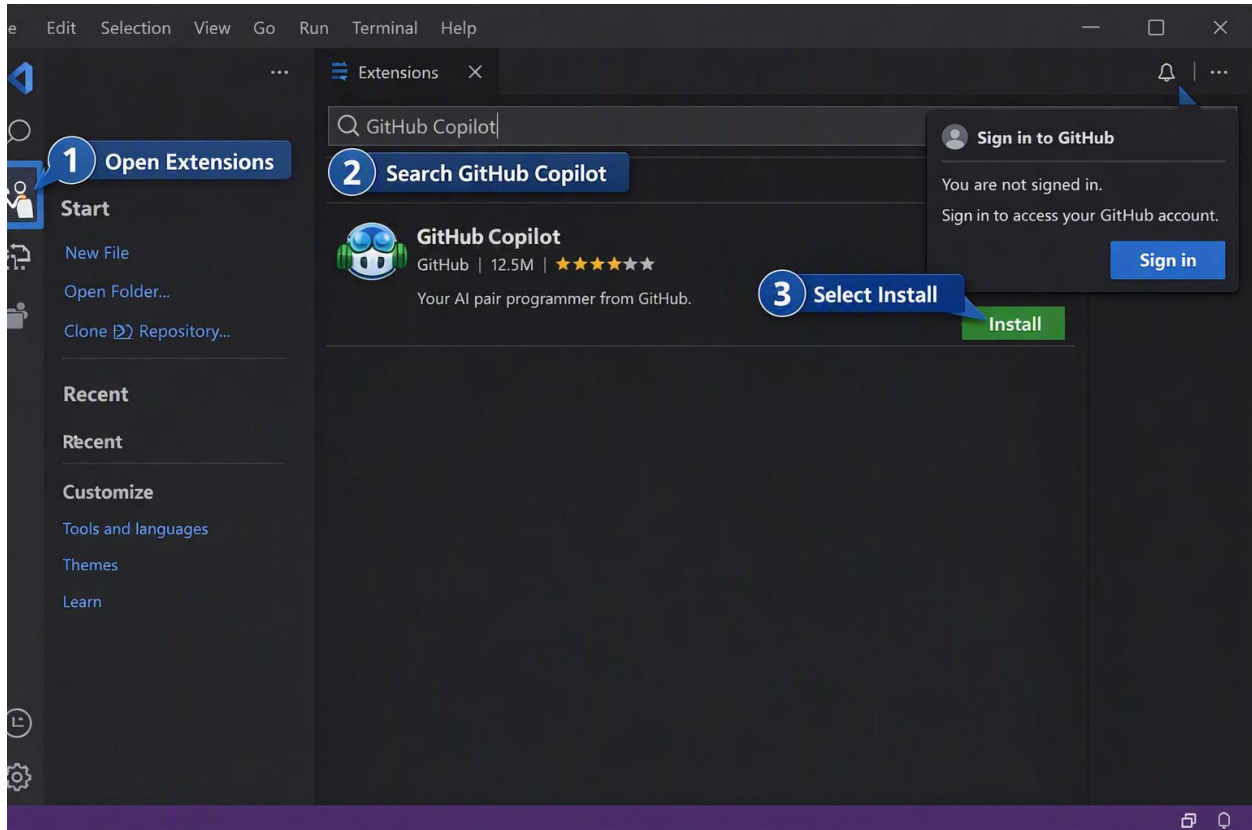
You are done when: `/skills` lists Power BI skills such as semantic-model authoring and report authoring, and Copilot reports no marketplace or authentication error.

3. Path B — Set up VS Code, GitHub Copilot, and Power BI Skills

Required report format: Before updating a Power BI report with Power BI Skills in GitHub Copilot CLI or VS Code, save the report as a **Power BI Project (PBIP)**. In Power BI Desktop, use **File > Save As** and select **Power BI Project (*.pbip)**. Then open the folder containing

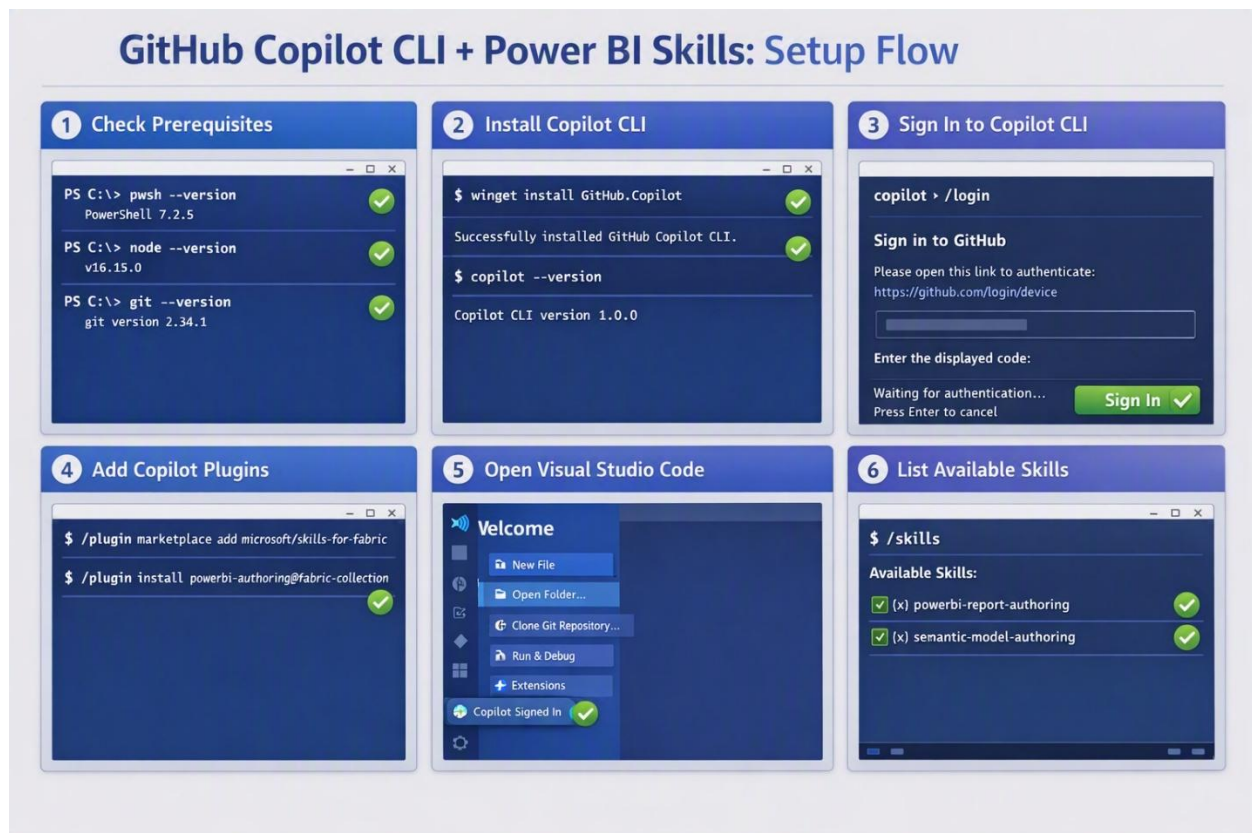
the .pbip file in VS Code or start GitHub Copilot CLI from that folder. Do not point the authoring workflow only at a .pbix file.

1. **Install and open VS Code.** On Windows, use the User Installer or the WinGet command above. On macOS, drag Visual Studio Code to Applications.
2. **Sign in to Copilot.** Hover over the Copilot icon in the status bar, select *Use AI Features*, and complete GitHub sign-in. Alternatively, open Extensions, search for the official GitHub Copilot extension, and select Install.



Screenshot 5. Install the official GitHub Copilot extension in VS Code and sign in with the licensed GitHub account.

3. **Open a trusted working folder.** Use *File > Open Folder*. For a Power BI project, open the folder that contains the .pbip file. Only trust folders from known sources.
4. **Open the integrated terminal.** Select *Terminal > New Terminal*. Start Copilot with `copilot`.
5. **Connect CLI and VS Code.** When Copilot CLI starts from the same open folder, it can connect automatically. If it does not, enter `/ide` and select the open VS Code workspace.
6. **Confirm the skills.** Enter `/skills`. The Power BI skills installed through the CLI should be available to the connected workflow.



Screenshot 6. Open the PBIP folder, start the integrated terminal, connect Copilot CLI to VS Code, and verify the available Power BI skills.

Optional — remote Power BI MCP for querying semantic models

The remote Power BI MCP server is a separate preview capability for natural-language queries against semantic models. It requires your Power BI administrator to enable the tenant setting, VS Code with GitHub Copilot, and Build permission on a semantic model. Use the Microsoft Learn one-click installer when available, or configure the endpoint <https://api.fabric.microsoft.com/v1/mcp/powerbi>. This is not required for basic report authoring skills.

4. First safe test

Paste this prompt first:

List the installed Fabric and Power BI skills. Explain which skill you would use to inspect this folder. Do not modify files, connect to a tenant, or run deployment commands.

Then try a read-only PBIP prompt: First confirm the report is saved as a Power BI Project and that the working folder contains the .pbip file. Then enter: “Inspect this Power BI

project and summarize its report pages, visuals, measures, and likely design issues. Show a plan only; do not change files.” After reviewing the plan, use a low-risk copy of the PBIP project for your first edit.

5. What the Power BI authoring skills can do

Goal	Use	Example prompt
Create or change report pages	Power BI Report Authoring skill	“Add a Sales Overview page with KPI cards, a monthly trend, a region slicer, and a product table.”
Improve layout and visual design	Report Design / Planner companion skills	“Produce a design brief first, using a 16:9 executive layout and accessible colors.”
Create or change model objects	Semantic-model skills and Power BI Modeling MCP	“Inspect the model and propose measures for margin, YTD sales, and prior-year variance.”
Validate the result	Desktop bridge and authoring tools	“Reload the PBIP, capture the rendered page, identify visual errors, and propose fixes.”

6. Troubleshooting

Symptom	Likely cause	Action
copilot is not recognized	CLI is missing or the terminal has an old PATH	Reinstall with WinGet or npm, close all terminals, then reopen PowerShell 7.
Node version error	Node.js is older than 22	Install the current Node.js LTS, reopen the terminal, and run node --version.
Sign-in succeeds but Copilot is unavailable	Wrong GitHub account or organization policy	Run /logout, then /login with the licensed account; ask the administrator to verify CLI policy.
Marketplace or plugin not found	Typing commands outside the interactive CLI,	Start copilot, rerun the slash commands, quit, restart, and check /skills.

	connectivity, or stale session	
VS Code does not connect	Folder mismatch or workspace not trusted	Open the same folder in VS Code, trust it, start Copilot CLI from that folder, or use <code>/ide</code> .
Agent cannot access Power BI content	Missing permissions, admin setting, or authentication	Confirm workspace and semantic-model permissions; for remote MCP, confirm the preview tenant setting.

7. Safety and governance checklist

- Save the report as a **Power BI Project (*.pbip)** before using Power BI Skills in GitHub Copilot CLI or VS Code; start with a copied PBIP project, a development workspace, or another low-risk environment.
- Ask for a plan before allowing file or tenant changes.
- Review generated PBIR, TMDL, DAX, data-source connections, and deployment targets.
- Do not paste passwords, access tokens, secrets, or customer-sensitive data into prompts.
- Keep existing workspace permissions, source control, approval, and deployment-pipeline controls in place.
- Commit a known-good baseline before agent edits so changes can be reviewed and reversed.
- Treat screenshots and generated visuals as validation evidence—not proof of semantic correctness.

8. Authoritative resources

- [Microsoft Fabric Community — AI-Powered Power BI reporting: From design to deployment with agent skills](#)
- [Microsoft Learn — Power BI Agentic overview](#)
- [Microsoft Learn — Install Skills for Fabric](#)
- [Microsoft Learn — Power BI Report Authoring skill](#)
- [GitHub Docs — Install GitHub Copilot CLI](#)
- [Visual Studio Code — Set up GitHub Copilot in VS Code](#)
- [GitHub Docs — Connect Copilot CLI to VS Code](#)

- [Microsoft Learn — Remote Power BI MCP server](#)